Low-Cost Optical Sensor System for Microplastic Detection in Environmental Water Samples

A MINI PROJECT II REPORT

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IN

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

KONGU ENGINEERING COLLEGE

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APRIL 2026

BONAFIED CERTIFICATE

This is to certify that the Mini Project - II report entitled **Low-Cost Optical Sensor System for Microplastic Detection in Environmental Water Samples** is the Bonafide record of project work done by **SANJEEVRAJ BOMMANNAN M 24ECR174, NANDITA M K 24ECR137 , SANJAY B 24ECR172** in partial fulfilment of the requirements for the award of the Degree of Bachelor of Engineering in **ELECTRONICS AND COMMUNICATION** of Anna university Chennai during the year 2025 - 2026.

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DECLARATION

We affirm that the Mini Project - II Report titled **Low-Cost Optical Sensor System for Microplastic Detection in Environmental Water Samples** being submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Microplastic contamination in water samples has emerged as a serious environmental and public health concern due to the widespread use of plastics and their gradual degradation into particles smaller than 5 mm. These particles enter water bodies through domestic waste, industrial discharge, and environmental breakdown of plastic materials, posing risks to aquatic ecosystems and human health. Conventional detection techniques such as Raman spectroscopy, FTIR analysis, and fluorescence imaging provide high accuracy but are expensive, require complex sample preparation, and are not suitable for real-time field deployment. This project proposes a low-cost optical sensor system for microplastic detection using a multi-wavelength approach. The system employs ultraviolet (UV), visible (white), and near-infrared (IR) LEDs to illuminate the sample, while a photodiode sensor measures variations in transmitted and scattered light intensity. The signals are processed using a signal conditioning circuit and an embedded microcontroller (ESP32) to enable real-time analysis and feature extraction. Experimental evaluation was carried out using three conditions: empty container, pure water, and microplastic-contaminated water samples. The results showed that pure water produced stable and low scattering signals, whereas contaminated samples exhibited noticeable intensity variations across different wavelengths, confirming the presence of microplastics. The system also demonstrated the ability to estimate relative concentration based on signal deviation. In conclusion, the proposed system offers a cost-effective, portable, and efficient solution for preliminary microplastic detection in water samples, making it suitable for on-site environmental monitoring and practical applications, although further improvements are required for accurate polymer type identification.

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SUSTAINABLE DEVELOPMENT GOALS

This project on microplastic detection in water using a low-cost optical sensor system directly supports United Nations Sustainable Development Goal 6. Microplastics, which are plastic particles smaller than 5 mm, are increasingly present in water sources due to plastic waste degradation and human activities, posing risks to both human health and aquatic life . Detecting these particles is challenging with conventional methods, as they require expensive laboratory equipment and are not suitable for continuous monitoring. The proposed system provides an affordable and portable solution for real-time detection of microplastics using optical sensing techniques. By measuring light scattering and intensity variations, it can identify contamination in water quickly and efficiently. This enables early detection, which is crucial for preventing long-term exposure and maintaining safe water quality. Additionally, the system can be deployed in various environments such as water treatment plants, lakes, and rural water sources, making it accessible even in resource-limited areas. By facilitating regular monitoring and providing immediate data, the project supports better water management and pollution control. Overall, this work contributes to ensuring the availability of clean and safe water, aligning with global efforts toward sustainable water management.

OBJECTIVE	**To develop a low-cost optical sensor system for real-time detection and estimation of microplastic contamination in water**

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CHAPTER – 1

INTRODUCTION

1.1 INTRODUCTION

Microplastics are plastic particles smaller than 5 mm that are increasingly found in oceans, rivers, groundwater, and drinking water sources. These particles are mainly produced from the degradation of plastic waste, industrial discharge, synthetic textiles, and domestic products. Due to their very small size, microplastics cannot be easily removed by conventional filtration systems and may enter the food chain, posing serious risks to aquatic organisms and human health.

Traditional microplastic detection methods such as Raman spectroscopy, FTIR spectroscopy, and electron microscopy provide high accuracy but require expensive laboratory equipment, skilled operators, and long processing time. These limitations make them unsuitable for real-time and field applications.

Recent research focuses on optical sensing techniques because they provide a fast, non-destructive, and cost-effective solution. Optical detection works based on light interaction with particles such as absorption, scattering, and transmission variation. When light passes through contaminated water, the intensity changes depending on the presence of particles.

This project proposes a Low-Cost Optical Sensor System for Microplastic Detection in Water using multi-wavelength LEDs (UV, Visible, and IR) and BPW34 photodiode sensors. The ESP32 microcontroller is used to process the optical signals and determine whether microplastics are present.

1.2 PROBLEM STATEMENT

Microplastic pollution is increasing rapidly due to excessive plastic usage and improper waste management. Monitoring microplastics is important, but existing detection methods have several drawbacks.

Major problems include:

• Laboratory-dependent detection methods such as Raman and FTIR spectroscopy are very expensive.

• Most existing systems cannot be used for field monitoring.

• Low-cost sensors like turbidity sensors cannot differentiate microplastics from dust or organic particles.

• Existing portable systems cannot estimate concentration accurately.

• No compact embedded system provides real-time optical analysis using low-cost components.

Because of these limitations, there is a strong need for a low-cost embedded microplastic detection system that can provide real-time monitoring using simple optical components.

This project aims to solve these problems by developing a compact optical sensing system using LEDs, photodiodes, and ESP32 processing.

1.3 OBJECTIVE

The main objective of this project is to develop a low-cost optical sensing system to detect microplastics in water using embedded hardware.

Specific objectives include:

• To design a low-cost optical detection system using LEDs and photodiodes.

• To detect microplastic presence using transmission and scattering principles.

• To compare pure water and contaminated water optical characteristics.

• To process sensor data using ESP32 microcontroller.

• To estimate concentration variation based on optical intensity difference.

• To develop a compact prototype for real-time monitoring.

• To create a portable environmental testing device.

CHAPTER – 2

LITERATURE REVIEW

2.1 INTRODUCTION

Microplastic detection has gained significant attention due to increasing environmental pollution and health risks. Various techniques such as spectroscopy, optical sensing, fluorescence imaging, and machine learning-based detection have been proposed by researchers. Each method provides advantages in detection accuracy, automation, or real-time monitoring but also suffers from limitations such as high cost, complex setup, or lack of portability.

Traditional laboratory techniques such as Raman spectroscopy provide accurate polymer identification but require expensive instruments and controlled environments. Recent studies focus on low-cost optical detection using LEDs and photodiodes for embedded implementation.

This chapter reviews important research works related to microplastic detection and discusses their advantages and limitations.

2.2 OPTICAL SCATTERING BASED DETECTION

Zhang et al. developed an optical scattering-based detection system that measures the scattering intensity variations caused by microplastic particles [1]. Their system demonstrated high sensitivity and accurate detection capability.

However, the system required expensive optical components and laboratory calibration, limiting its portability. This study shows that optical scattering is effective but requires cost optimization.

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2.3 EMBEDDED OPTICAL MONITORING SYSTEMS

Kumar and Lee proposed an embedded real-time microplastic monitoring system using LED and photodiode sensors [2]. Their system provided portable detection and low power consumption.

However, the system could only detect the presence of particles and could not accurately classify polymer types. This shows that embedded optical systems are suitable for low-cost detection but require further improvements.

2.4 IoT BASED MICROPLASTIC MONITORING

Sharma et al. developed an IoT-based optical water quality monitoring system including microplastic detection [3]. The system enabled real-time data transmission and monitoring.

Although the system provided continuous monitoring, it lacked high selectivity and required internet connectivity. This indicates the need for simpler standalone embedded systems.

2.5 MACHINE LEARNING BASED DETECTION

Cerioli et al. developed a TinyML-based optical classification system to detect microplastics using embedded machine learning [4]. Their system achieved low power processing and improved detection accuracy.

However, the system required trained datasets and complex model development. This shows the potential of ML but also highlights implementation complexity.

Similarly, Patel et al. proposed a deep learning-based microscopic classification system [8]. Their system achieved good classification accuracy but required high computational resources.

2.6 MICROWAVE AND SPECTROSCOPY METHODS

Zhao et al. developed a microwave resonator-based microplastic detection system capable of concentration estimation [5]. However, the system could not identify polymer types.

Liu et al. used Raman spectroscopy for polymer identification [6]. Their system achieved high accuracy but required expensive instrumentation.

Lu et al. proposed a portable SERS-based nanoplastic detection system capable of detecting nano-scale plastics [7]. However, the system required costly substrates.

These studies show that spectroscopy methods provide high accuracy but are expensive.

2.7 FLUORESCENCE AND IMAGING METHODS

Vitali et al. developed a fluorescence imaging system combined with AI for microplastic quantification [9]. The system provided good concentration estimation but required chemical staining.

Passananti et al. proposed a flow cytometry-based rapid microplastic analysis method [10]. The system provided fast detection but had limited field deployment capability.

Kim et al. proposed UV-induced fluorescence detection of microplastics [11]. The system showed polymer sensitivity but required controlled testing conditions.

2.8 ADVANCED OPTICAL SENSOR METHODS

Xiong et al. developed an optical fiber-based nanosensor for detecting plastic particles [12]. The system detected nanoplastics but was limited to specific polymer types.

Garcia et al. developed a portable microplastic analyzer using optical imaging [13]. The system provided moderate accuracy but required image processing.

Brown et al. proposed multi-spectral optical fingerprinting for microplastic detection [14]. Their system provided spectral analysis but required complex hardware.

Ahmed et al. developed a laser-induced spectral analysis method for microplastic detection [15]. The system provided high accuracy but required expensive laser setups.

2.9 RESEARCH GAP AND SUMMARY

The literature review indicates that numerous techniques have been developed for the detection of microplastics using optical sensing, spectroscopy, imaging, and machine learning approaches. Each of these methods offers specific advantages but also suffers from limitations related to cost, portability, system complexity, and real-time implementation. Spectroscopy-based techniques such as Raman and Surface-Enhanced Raman Scattering (SERS) provide high accuracy in polymer identification; however, they require expensive laboratory instruments and skilled operation. Optical LED–photodiode systems offer a low-cost and portable alternative, but they have limited capability in polymer classification. IoT-based monitoring systems enable real-time data acquisition and remote monitoring, but they increase system complexity and rely on continuous internet connectivity. Machine learning-based approaches improve detection and classification performance, yet they require large training datasets and higher computational resources. Fluorescence and imaging-based methods provide good concentration estimation, but they involve chemical preparation and complex image processing techniques. Advanced optical methods such as laser-based spectral analysis achieve high precision, but significantly increase the overall system cost. From this analysis, it is evident that no existing system effectively combines low-cost hardware implementation, portability, multi-wavelength optical detection, real-time analysis capability, and a simple compact design. Therefore, a clear research gap exists for developing an affordable, portable, and real-time microplastic detection system using simple optical components and embedded processing. The proposed project addresses this gap by utilizing multi-wavelength LEDs (UV, visible, and IR), BPW34 photodiode sensors, and an ESP32 microcontroller for signal processing. The system detects microplastics based on variations in light transmission and scattering while maintaining low cost and ease of use. This chapter has reviewed existing techniques and highlighted the necessity for an economical embedded optical detection system, and the subsequent chapter presents the existing methods and the proposed methodology in detail.

CHAPTER – 3

EXISTING METHODOLOGY

3.1 INTRODUCTION

Microplastic detection is commonly performed using spectroscopy, optical imaging, fluorescence analysis, and optical scattering techniques. These methods mainly focus on identifying plastic particles based on their chemical, optical, and physical properties. Many research works have demonstrated high detection accuracy using these techniques. However, most of these systems require expensive laboratory equipment and complex processing. This section discusses some important existing methods along with their advantages and limitations based on previous research papers.

3.2 RAMAN SPECTROSCOPY METHOD

Liu et al. proposed a Raman spectroscopy-based method for identifying polymer types of microplastics by analyzing molecular vibration signatures [6]. This method uses laser excitation to obtain spectral fingerprints of plastic materials, enabling accurate identification of different polymer types.

Advantages: The method offers several advantages, including high accuracy in polymer identification, non-destructive testing capability, the ability to detect very small particles, and reliable material classification.

Limitations: The method also has certain limitations, such as very expensive instrumentation, the requirement of a laboratory setup, the need for skilled operators, and its unsuitability for portable and field-based applications.

3.3 OPTICAL SCATTERING METHOD

Zhang et al. developed an optical scattering-based detection system where light scattering variation is used to detect microplastic particles [1]. The system measures scattering intensity changes when particles interact with light.

Advantages: The method provides high sensitivity detection and enables non-contact measurement, making it suitable for safe and efficient analysis. It offers a fast detection process and is effective for identifying the presence of particles in water samples.

Limitations: The method requires precise optical alignment for accurate results and involves relatively expensive optical components. It also requires proper calibration to maintain accuracy and has limited capability in identifying specific polymer types.

3.4 MACHINE LEARNING BASED OPTICAL DETECTION

Cerioli et al. proposed a TinyML-based optical signal classification system to detect microplastics using embedded machine learning [4]. The system processes optical signals and classifies microplastic presence using trained models.

Advantages: The method provides improved detection accuracy and supports low-power embedded processing, making it suitable for compact systems. It enables real-time classification of microplastics and offers a scalable approach for future enhancements and larger datasets.

Limitations: The method requires a properly prepared training dataset and involves complexity in model development. It also needs optimization for efficient performance and may produce limited or inaccurate results without sufficient and well-trained data.

3.5 FLUORESCENCE IMAGING METHOD

Vitali et al. proposed a fluorescence imaging system combined with artificial intelligence to quantify microplastic concentration [9]. The method uses fluorescent dyes to highlight microplastic particles under specific wavelength illumination.

Advantages: The method provides good concentration estimation and enhances the visibility of particles, making it suitable for detailed research analysis. It is also capable of detecting small particles with improved clarity.

Limitations: The method requires chemical staining for effective detection and depends on imaging equipment, which increases system requirements. It also involves complex processing, making it less suitable for simple and real-time applications.

3.6 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

Method	Methodology	Technology Used	Advantages	Limitations
Raman Spectroscopy Method [6][7]	Uses laser light interaction to identify molecular structure of plastics	Laser source, spectrometer, optical filters	Very high accuracy, polymer identification possible	Very expensive, laboratory dependent, complex operation
Optical Scattering Method [1][2]	Measures light scattering variation when particles interact with light	LED/Laser source, photodetector, optical setup	Fast detection, good sensitivity, non-destructive	Requires calibration, limited polymer classification
Machine Learning Based Optical Detection [4][8]	Uses optical signal data and ML algorithms for classification	Optical sensors, microcontroller, ML models	Improved detection accuracy, automation possible	Requires training data, complex model development
Fluorescence Imaging Method [9][13]	Uses fluorescent dyes and UV light to visualize microplastics	UV light source, microscope, imaging system	Good particle visibility, concentration estimation	Requires chemical staining, expensive imaging setup

Table 3.6.1

From this comparison, Raman spectroscopy provides high accuracy but is expensive and laboratory dependent. Optical scattering provides a simple detection method but lacks material classification capability. Machine learning-based detection improves accuracy but increases system complexity. Fluorescence imaging provides good visualization but requires chemical preparation. Therefore, there is a need for a low-cost, portable optical detection system using simple LEDs and photodiode sensors with embedded processing, which is addressed by the proposed system.

3.7 SUMMARY

This chapter discussed the existing methodologies used for microplastic detection such as Raman spectroscopy, optical scattering methods, machine learning-based optical detection, and fluorescence imaging techniques. Raman spectroscopy provides highly accurate polymer identification but requires expensive laboratory equipment. Optical scattering methods provide fast detection but have limited capability in material classification. Machine learning-based optical detection improves detection accuracy but requires training datasets and higher computational resources. Fluorescence imaging methods provide good particle visualization but require chemical staining and complex imaging setups.

From this study, it is observed that most existing systems either provide high accuracy with high cost or low cost with limited detection capability. None of the systems provide a balance between low cost, portability, and real-time detection using simple embedded hardware.

Therefore, there is a need for a compact, low-cost optical detection system that can detect microplastics using simple optical components and embedded processing. The next chapter explains the proposed methodology used in this project to overcome these limitations by using multi-wavelength LEDs, photodiode sensors, and ESP32 microcontroller processing.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 INTRODUCTION

The proposed system is designed to detect microplastics in water using optical sensing principles. The system uses a low-cost embedded approach instead of expensive laboratory techniques. The detection is based on the interaction between light and microplastic particles, where the presence of particles affects light transmission, absorption, and scattering characteristics. By measuring these optical variations, the system determines whether microplastics are present in the water sample.

The proposed system uses multi-wavelength LEDs such as UV, Visible (white), and IR LEDs as light sources and a BPW34 photodiode as the optical detector. The ESP32 microcontroller is used to process the sensor signals and determine the presence of microplastics. The system is designed to be compact, portable, and suitable for real-time monitoring.

4.2 SYSTEM ARCHITECTURE

The system consists of a multi-wavelength LED array, a water sample chamber, a photodiode sensor, a signal conditioning circuit, and an ESP32 microcontroller. The LEDs are used to illuminate the water sample from different wavelengths. The photodiode is placed to receive transmitted and scattered light from the sample. The photodiode converts light intensity into electrical signals.

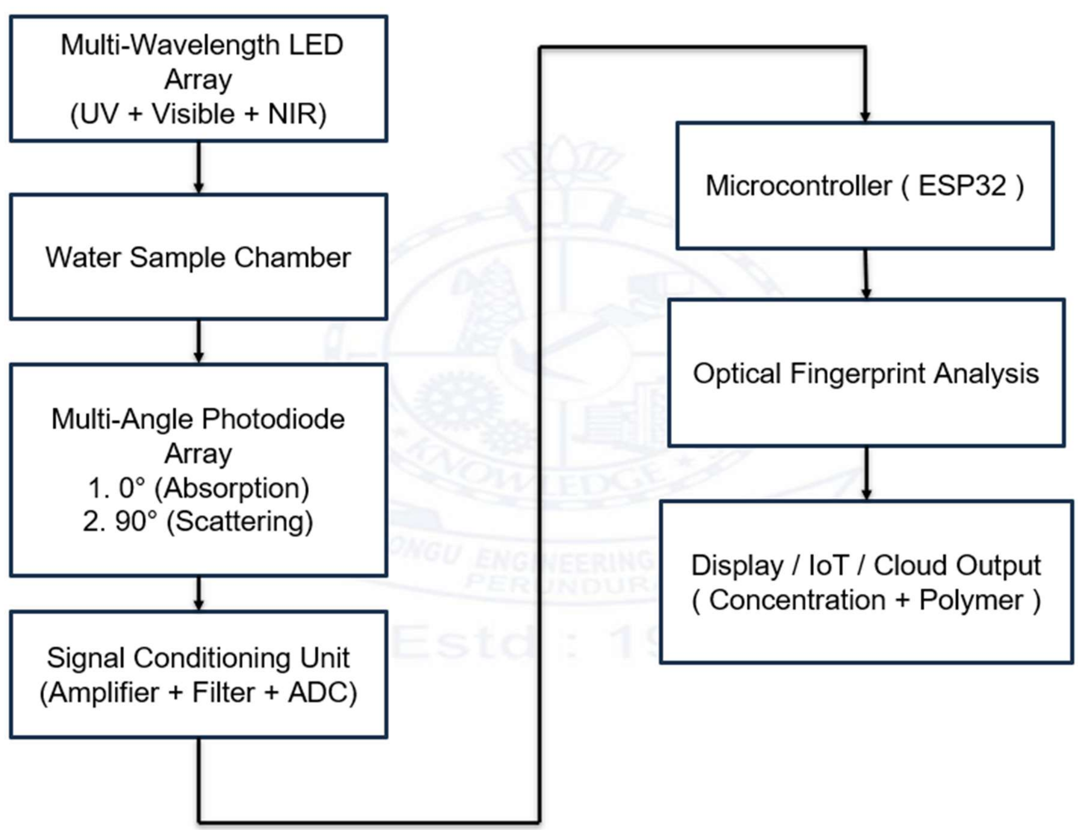
Since the photodiode output voltage is very small, an LM358 operational amplifier is used to amplify the signal. The amplified signal is then given to the ESP32 analog input pins. The ESP32 processes the sensor readings and compares them with reference values obtained from pure water. Based on the difference in readings, the system determines whether microplastics are present.

4.3 WORKING PRINCIPLE

The working principle of the proposed system is based on optical transmission and scattering measurement. When light passes through pure water, most of the light is transmitted because there are no particles to obstruct the path. When microplastics are present, they cause scattering and absorption of light, which reduces transmission intensity and increases scattered light intensity.

The system first measures the empty container reading to establish the baseline value. Then pure water is tested to obtain the reference reading. Finally, the contaminated water sample is tested. The ESP32 compares the pure water value and the sample value. If the difference exceeds a predefined threshold, the system indicates the presence of microplastics. This comparison method improves detection reliability.

4.4 BLOCK DIAGRAM



4.5 COMPONENTS USED

4.5.1 ESP32 Microcontroller

The ESP32 is the main processing unit of the system. It is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities. In this project, the ESP32 reads the photodiode sensor values, processes the data, and determines the presence of microplastics based on intensity variations.

Features and Specifications

- Dual-core 32-bit Xtensa LX6 microprocessor
- Operating frequency: 160 MHz to 240 MHz
- Built-in Wi-Fi 802.11 b/g/n and Bluetooth v4.2 (BR/EDR + BLE)
- Flash memory: 4 MB
- SRAM: 520 KB
- Operating voltage: 3.0 V to 3.6 V
- 30 to 36 GPIO pins
- ADC (12-bit) and DAC support
- UART, SPI, I2C, PWM interfaces available
- Power consumption (Active mode): < 240 mA

Figure 4.5.1**:** ESP32 Microcontroller

4.5.2 BPW34 PHOTODIODE

The BPW34 photodiode is used to detect the light intensity after interaction with the water sample. It converts the incident light into a proportional electrical current which is used to analyze transmission and scattering variations. Due to its wide spectral response range (400–1100 nm) and fast response time, it is suitable for optical sensing applications.

Features:

The BPW34 photodiode provides high sensitivity and low noise performance, making it suitable for accurate optical measurements. It also offers fast response and reliable detection, which improves measurement stability in the microplastic detection system.

Specifications:

Table 4.5.2

Parameter	**Specification**
Spectral Range	400–1100 nm
Peak Sensitivity	900 nm
Operating Voltage	0–60V
Response Time	Fast
Package Type	Through hole

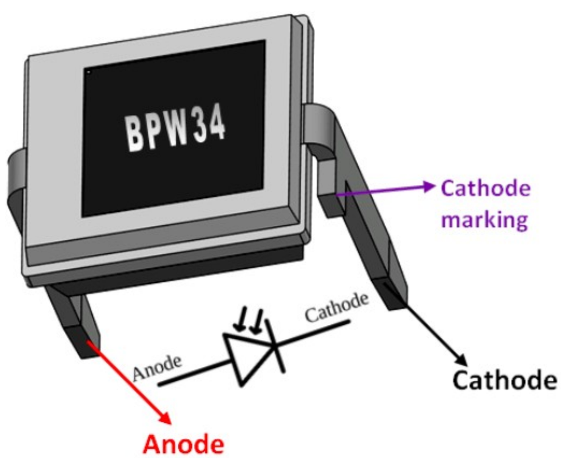


Figure 4.5.2**:** BPW34 PHOTO DIODE

4.5.3 UV LED

The UV LED is used to analyze fluorescence characteristics of microplastic particles. It emits ultraviolet light which interacts with plastic particles and helps identify optical variations.

Features

The UV LED provides stable ultraviolet output and low power consumption. It is suitable for fluorescence-based detection and improves optical analysis capability.

Specifications

Table 4.5.3

Parameter	**Specification**
Wavelength	365 nm
Forward Voltage	3–3.4V
Current Rating	20mA
LED Type	5mm

Figure 4.5.3: UV LED

4.5.4 IR LED

The IR LED is used to measure transmission characteristics of the water sample. It helps in identifying absorption variations caused by microplastic particles. The component offers good penetration capability and provides stable infrared emission, making it suitable for consistent optical measurements. It is a low-cost component, which supports economical system design, and is well suited for transmission-based measurement applications.

Specifications

Table 4.5.4

Parameter	Specification
Wavelength	**850 nm**
Forward Voltage	**1.2–1.5V**
Current	**20–50mA**
LED Type	**5mm**

Figure 4.5.4: IR LED

4.5.5 WHITE LED

The white LED is used to measure scattering effects in the water sample. When microplastics are present, scattering increases which affects detected intensity. The component provides high brightness output and ensures stable light emission for reliable performance. It also has a low power requirement, making it suitable for energy-efficient operation in embedded systems.

Specification:

Table 4.5.4

Parameter	**Specification**
Forward Voltage	**3V**
Current	**20mA**
Light Type	**Visible white**
LED Size	**5mm**

Figure 4.5.5: IR LED

4.5.6 LM358 OPERATIONAL AMPLIFIER

The LM358 operational amplifier is used to amplify the small electrical signal produced by the photodiode. Since the photodiode output is very weak, amplification is necessary before giving it to the ESP32 ADC. The component supports low power operation and is available in a dual amplifier package, making it efficient and compact for circuit design. It provides stable signal amplification and operates over a wide voltage range, ensuring reliable performance under varying conditions. Additionally, it is well suited for sensor signal conditioning in embedded and measurement systems.

Specifications:

Table 4.5.6

Parameter	**Specification**
Supply Voltage	3–32V
Number of Op-Amps	2
Gain Bandwidth	1 MHz
Input Offset Voltage	Low

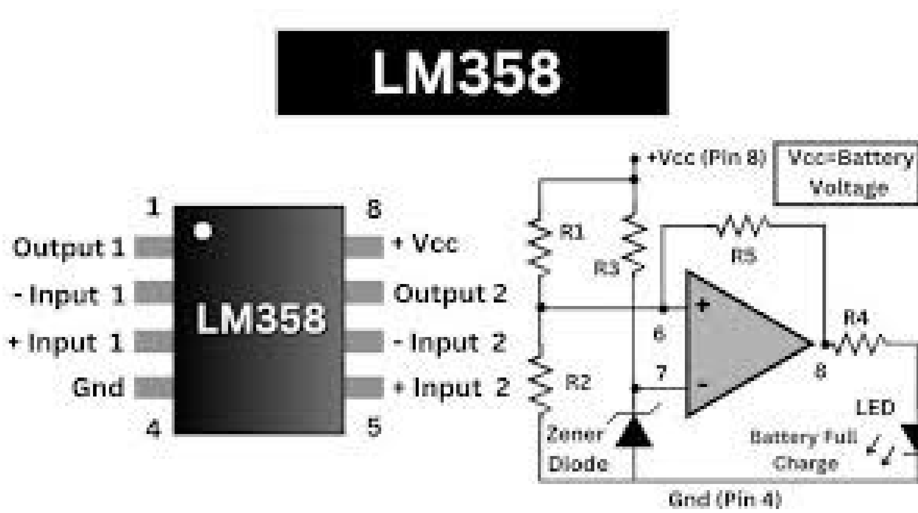


Figure 4.5.6: LM358

4.5.7 WATER SAMPLE CHAMBER

The water sample chamber is used to hold the test sample during optical testing. It is made of transparent acrylic or glass to allow proper light transmission. The component offers high optical transparency, allowing accurate transmission of light during measurements. It is a low-cost container that supports economical system design and enables easy sample handling during experiments. The structure is reusable, making it practical for repeated testing, and it is well suited for various optical experiments.

Specification:

Table 4.5.7

Parameter	**Specification**
Material	**Acrylic / Glass**
Width	**~6–10 cm**
Transparency	**High**
Type	**Static container**

Figure 4.5.7: LM358

4.6 COST ANALYSIS

Table 4.6.1

S.No	Component	Specification	Purpose	Approx Cost (₹)
1	UV LED	365 nm, 5mm	Fluorescence testing	8
2	White LED	5mm, 20mA	Scattering measurement	6
3	IR LED	850 nm	Transmission analysis	8
4	BPW34 Photodiode	400–1100 nm	Optical detection	100
5	LM358 Op-Amp	Dual Op-Amp	Signal amplification	15
6	ESP32	Dual core MCU	Processing unit	300
7	Sample Chamber	Acrylic container	Water testing	50
8	Breadboard & Wires	Standard	Circuit setup	40
9	Passive Components	Resistors, capacitors	Circuit support	20

Total Estimated Cost ≈ ₹550
The cost may vary depending on component availability and quality.

4.7 ADVANTAGES OF THE PROPOSED SYSTEM

• Low-cost system compared to Raman and FTIR spectroscopy methods

• Portable design suitable for field testing and environmental monitoring

• Real-time detection without laboratory dependency

• Simple hardware implementation using LEDs and photodiode

• Easy calibration and maintenance

• No complex sample preparation required

4.8 SUMMARY

This chapter explained the proposed methodology, system architecture, working principle, and hardware components used in the microplastic detection system. The specifications and features of the components were discussed along with the cost analysis of the prototype. The proposed system provides a simple, economical, and portable solution for detecting microplastics using optical sensing techniques.

CHAPTER – 5
RESULT AND DISCUSSION

5.1 RESULT

Empty Container (Baseline Condition)

The system was first tested using an empty sample container to establish a reference condition. In this case, there are no particles or liquid present to interact with the incident light.
As observed in the output image, the photodiode readings show a baseline signal level. This represents system noise and direct light reception without any scattering or absorption.
This step is important because it ensures that the system is properly calibrated and that any further changes in readings are due to the sample, not environmental or hardware errors.

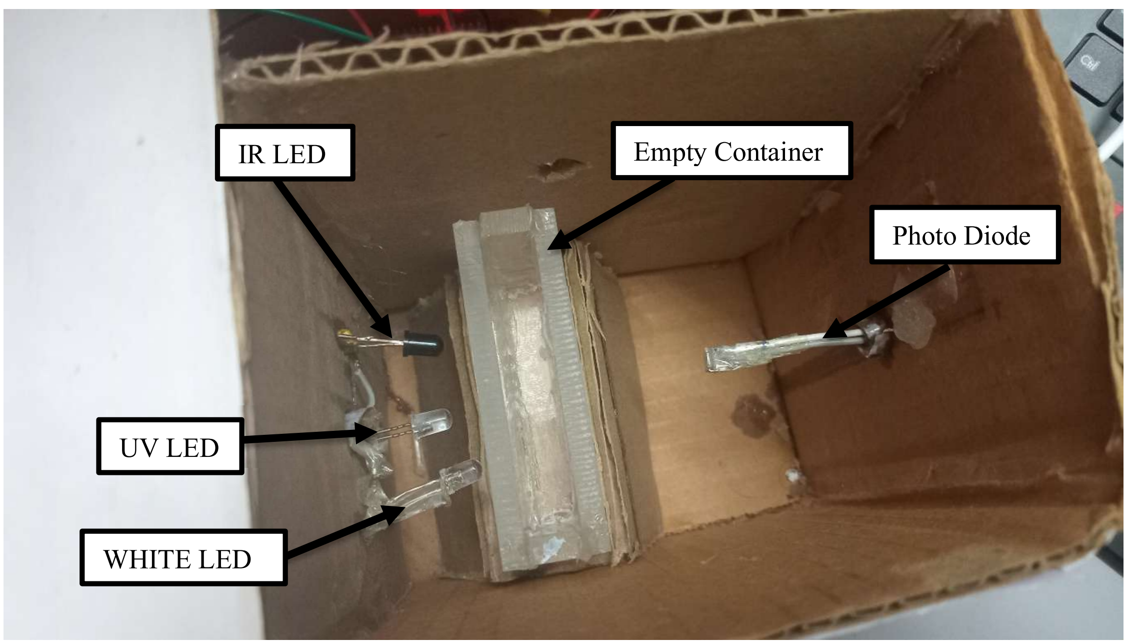


Figure 5.1.1: Empty Container (Baseline Measurement Condition)

Pure Water Sample

Next, the container was filled with distilled water and tested under the same conditions. Water is mostly transparent, so light passes through with minimal disturbance. From the corresponding output image, the readings show a slight variation compared to the empty container, mainly due to minor reflection and refraction within the water. However, there is no significant increase in scattering. This confirms that the system correctly identifies clean water and does not falsely detect microplastics.

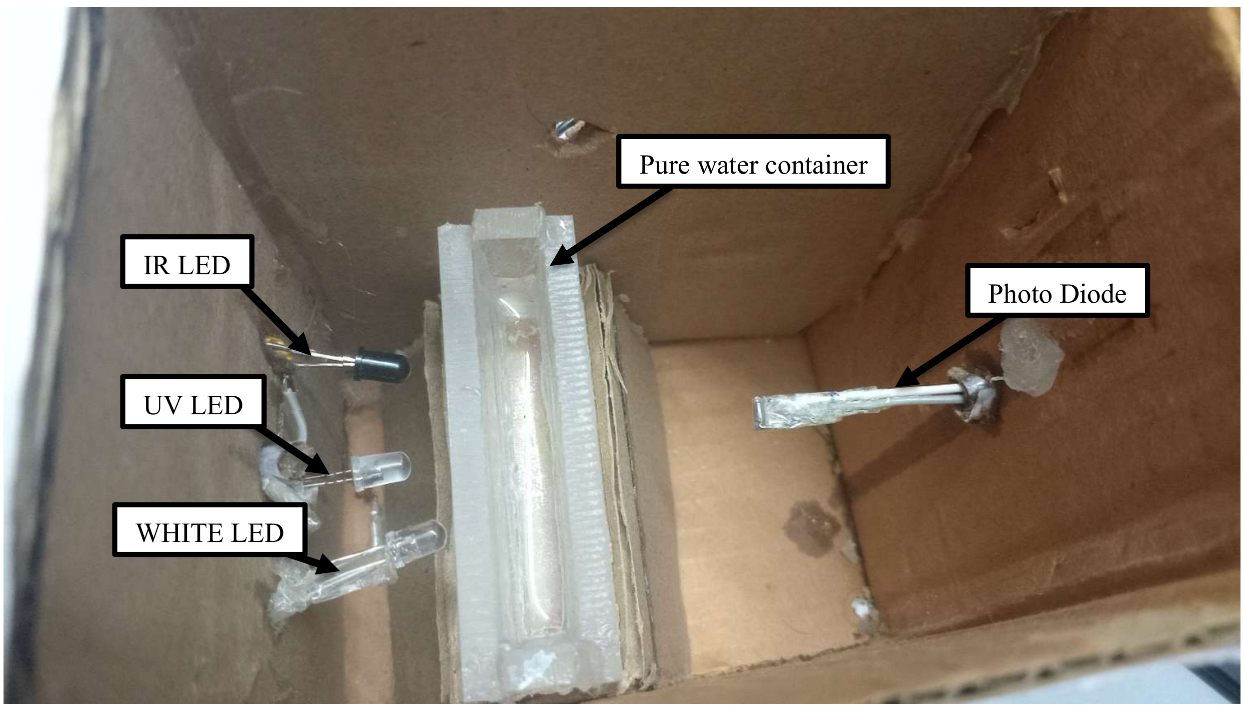


Figure 5.1.2: Pure Water Container (Distilled Water)

Microplastic Contaminated Sample

Finally, a water sample containing microplastic particles was tested under the same experimental conditions. The sample consisted of water mixed with small plastic particles of approximate size range 100 µm to 1 mm, representing typical microplastic contamination. The concentration of microplastics was maintained at a measurable level to observe clear optical variations. In this case, the suspended particles interact strongly with the incident light. The output readings show a significant increase in scattering values and a reduction in transmission intensity. This occurs because microplastic particles scatter light in multiple directions and reduce the amount of light passing directly through the sample. The difference between this reading and the previous two conditions is clearly distinguishable, enabling the system to effectively detect the presence of microplastic contamination.

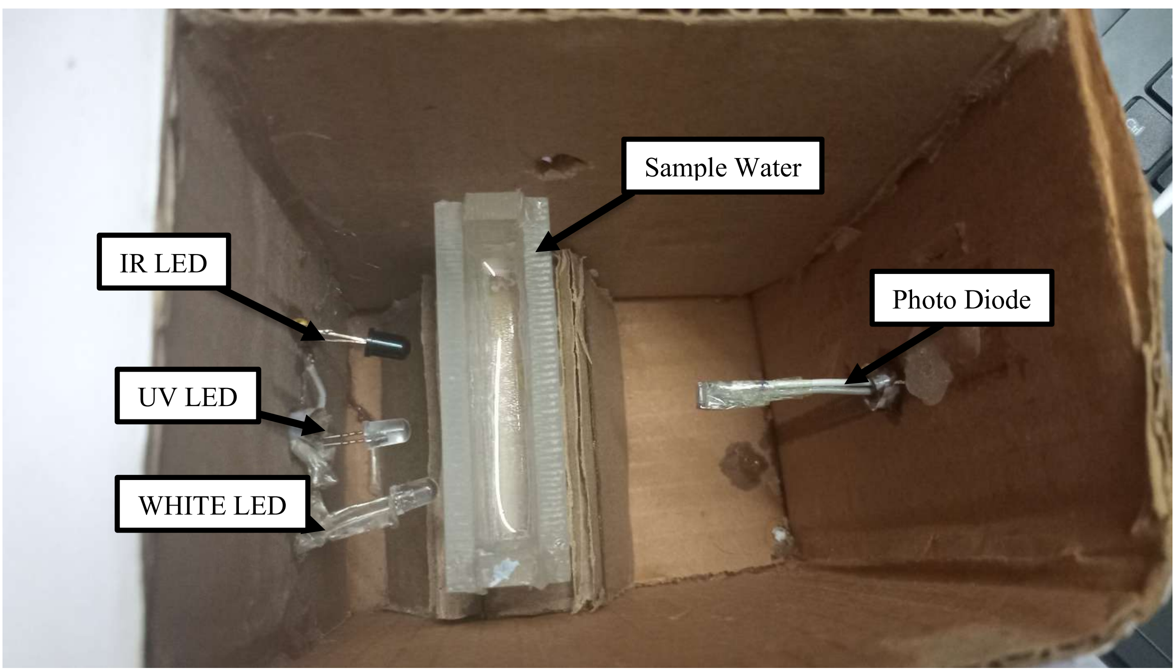


Figure 5.1.3: Sample Water Container (Microplastic Contaminated Sample)

Output Display and Detection Result

The final image shows the system output displayed through the serial monitor. The ESP32 processes the sensor data in real time and compares it with reference values.

Based on the observed variations, the system successfully classifies the sample as:

- Reference (Empty)
- Clean Water
- Microplastic Detected

This confirms that the system performs real-time detection and decision-making using optical sensing.

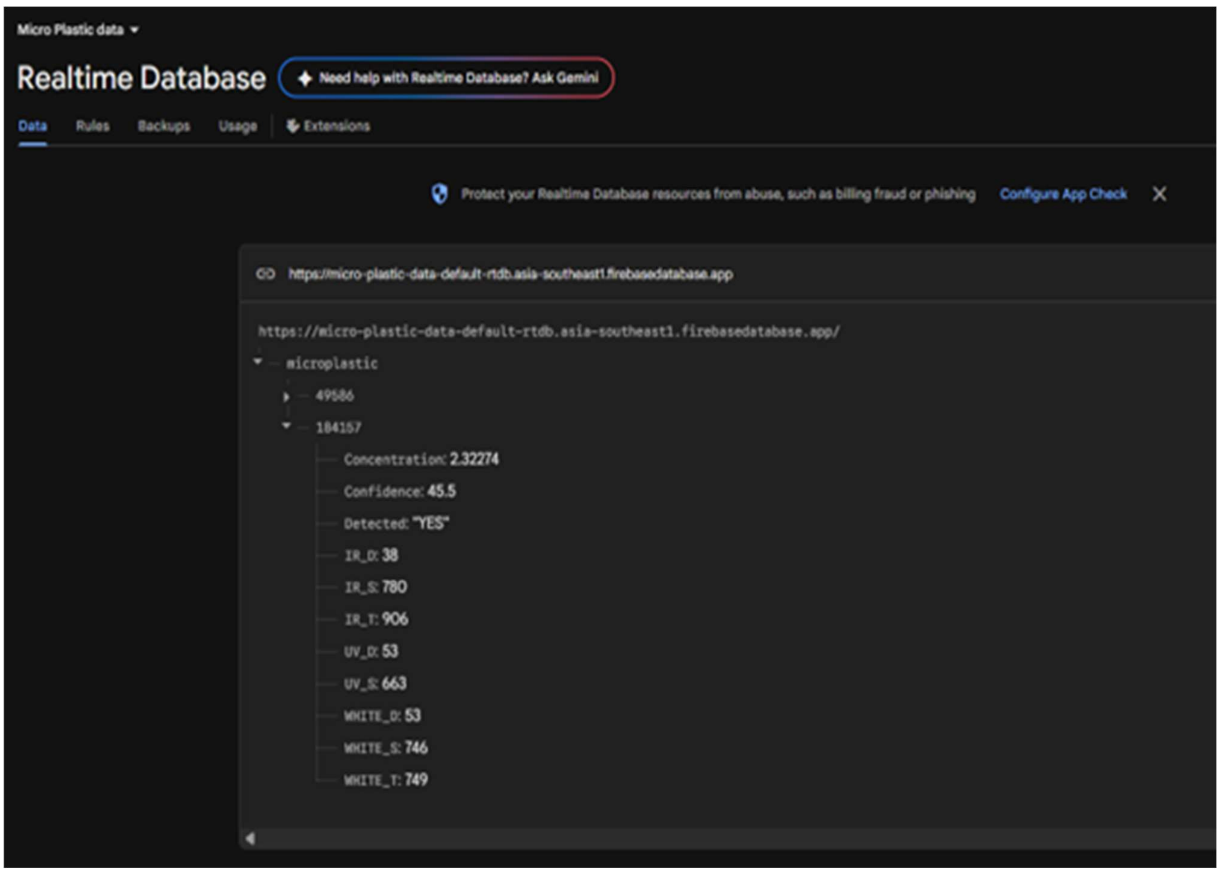


Figure 5.1.4: Realtime Database Output (Microplastic Detection Results)

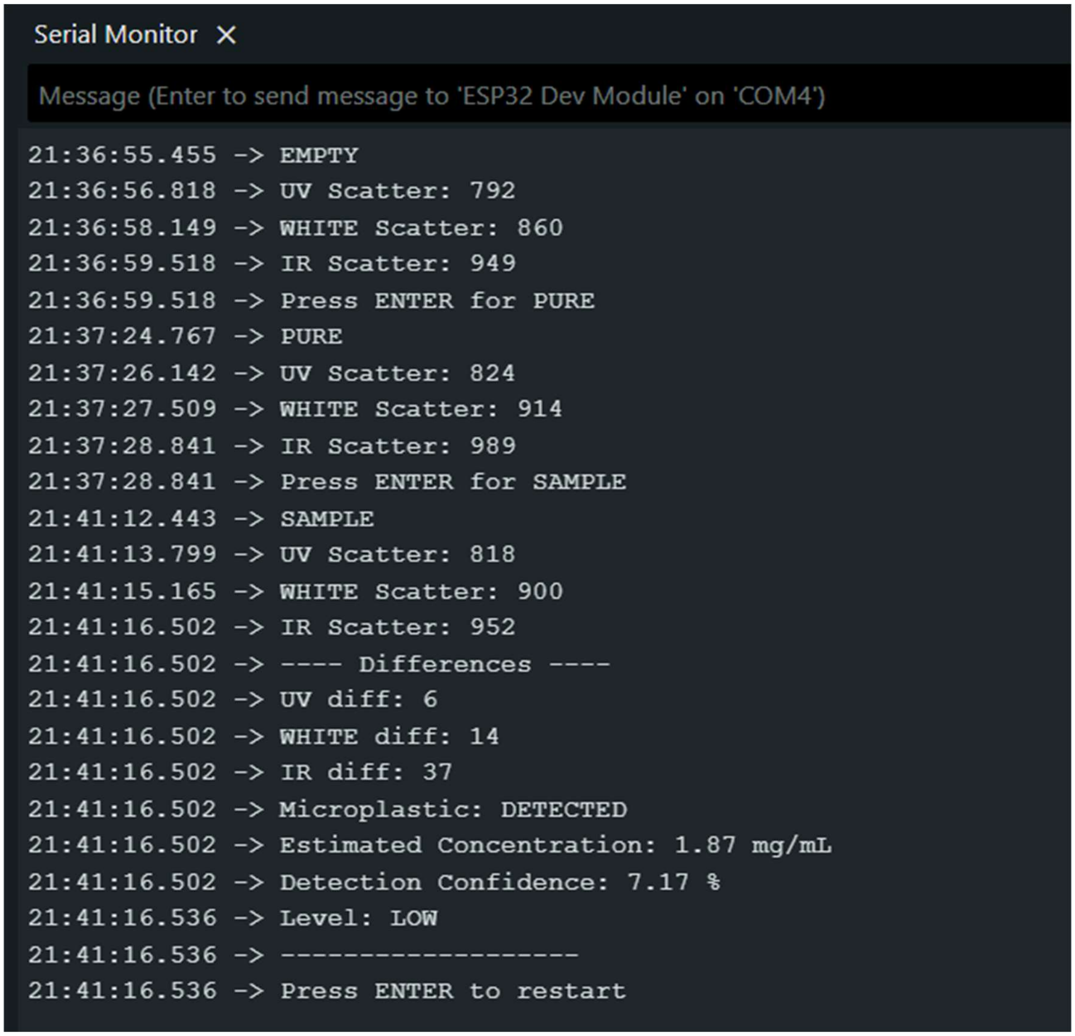


Figure 5.1.5: Serial Monitor Output

5.2 DISCUSSION

From the experimental results, it is observed that the optical characteristics of the sample vary significantly under different test conditions. The empty container provides a baseline with maximum transmission and no scattering. In the case of pure water, the transmission remains high with only slight variations due to minor reflection and refraction, and negligible scattering is observed. However, for the microplastic-contaminated sample, there is a clear reduction in transmission and a significant increase in scattering values, indicating the presence of suspended particles. The UV LED shows noticeable variation due to fluorescence interaction, while the white LED exhibits increased scattering, and the IR LED helps in identifying transmission reduction.

Table 5.1: Comparison of Optical Sensor Results for Different Water Samples

Condition	**Transmission**	**Scattering**	**Concentration (mg/mL)**	**Result**
Empty	Maximum	None	0	No microplastics
Pure Water	High	Very Low	~0	Clean water
Sample Water	Reduced	High	~2.3	Microplastics detected

From the above comparison, it is evident that the system can clearly differentiate between clean and contaminated water samples based on optical intensity variations. The increase in scattering and decrease in transmission in the sample water confirms the presence of microplastics. The system demonstrates reliable detection capability using simple optical sensing techniques. However, the performance depends on proper alignment of LEDs and photodiode, as well as stable sample placement during testing.

5.3 IMPACT AND INNOVATION

The proposed system provides a low-cost alternative to expensive laboratory detection methods. It demonstrates how embedded systems and optical sensing can be used for environmental monitoring. The project shows innovation by combining multi-wavelength optical sensing with embedded processing in a compact design.

The system can be further improved by integrating machine learning algorithms for polymer classification and concentration estimation. Future improvements may also include IoT connectivity for remote monitoring and data logging. The project demonstrates a practical approach toward developing affordable environmental sensing devices.

5.4 SUMMARY

The experimental results confirm that the proposed optical sensing system can detect variations between pure water and contaminated water samples. The system successfully demonstrated real-time detection using simple hardware components. The next chapter presents the conclusion and future scope of the project.

CONCLUSION AND FUTURE SCOPE

Conclusion

The proposed low-cost optical sensor system for microplastic detection in water samples was successfully designed, developed, and tested using simple and efficient hardware components. The system operates based on optical sensing principles, specifically transmission reduction and scattering variation, to identify the presence of microplastic particles. Experimental evaluation using empty, pure, and microplastic-contaminated water samples demonstrated that the system can effectively distinguish between clean and contaminated conditions through measurable intensity differences. The results showed a clear increase in scattering and a decrease in transmission in the presence of microplastics, confirming the reliability of the approach. The use of cost-effective components such as UV, white, and IR LEDs, BPW34 photodiode, LM358 amplifier, and ESP32 microcontroller ensures that the system remains economical, portable, and suitable for real-time applications. Compared to conventional laboratory-based techniques like Raman spectroscopy and FTIR, which are expensive and complex, the proposed system offers a practical alternative for preliminary detection and field-level monitoring. Although the system currently focuses on detecting the presence and relative concentration of microplastics, it provides a strong foundation for further enhancements such as polymer classification and improved accuracy. Overall, the project demonstrates that optical sensing combined with embedded processing can be effectively utilized to develop a compact, low-cost, and real-time microplastic detection system for environmental monitoring and water quality assessment applications.

Future Scope

The proposed system can be further enhanced to improve its performance and functionality. Machine learning algorithms can be incorporated for polymer type classification and improved detection intelligence, while concentration estimation can be refined using proper calibration techniques. A multi-angle photodiode arrangement can enhance detection accuracy by capturing detailed scattering information, and a compact PCB design can improve portability for field applications. Mobile application integration can enable real-time monitoring and user-friendly data visualization. Additionally, optical filters can improve wavelength isolation and measurement precision, and AI-based analysis can enhance detection reliability. Overall, these improvements can make the system more robust, efficient, and suitable for real-world microplastic detection applications.

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